

# SkyBook

An airline reservations platform, built end to end

Recruiter edition — the 12-page summary of a 68-page engineering case study

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## At a glance

| Item         | Detail                                                                                                                                                                      |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| What         | A working airline: anonymous search, real seat maps, payment, confirmation email, check-in, scannable boarding pass, cancellations with fare-rule refunds, an admin console |
| Shape        | Eight Spring Boot applications (one API gateway, seven domain services), React 19 web client, PostgreSQL, Kafka, full observability                                         |
| Built        | 35 calendar days of designed increments, then a measured quality and release-engineering revision                                                                           |
| Quality      | SonarCloud gate PASSING: 91.2% coverage, 0 bugs, 0 vulnerabilities, security and reliability rating A                                                                       |
| Tests        | 1,722 backend executions + 55 frontend + a 24-test end-to-end certification that races real buyers for one seat - zero failures at the evidence commit                      |
| Environments | A real promotion ladder: LOCAL > DEV > SIT > QA > PERF > UAT > STAGING > PROD, plus a weekly disaster-recovery drill that restores and verifies row counts                  |
| Deployment   | One cloud VM, 18 containers, automatic TLS - promoted by image digest, never rebuilt on the way to production                                                               |
| Author       | Praveenreddy Somireddy - solo: requirements, architecture, backend, frontend, testing, CI/CD, cloud deployment and documentation, every decision and merge owned end to end |

[Repository: github.com/praveencloudlab/skybook](https://github.com/praveencloudlab/skybook)

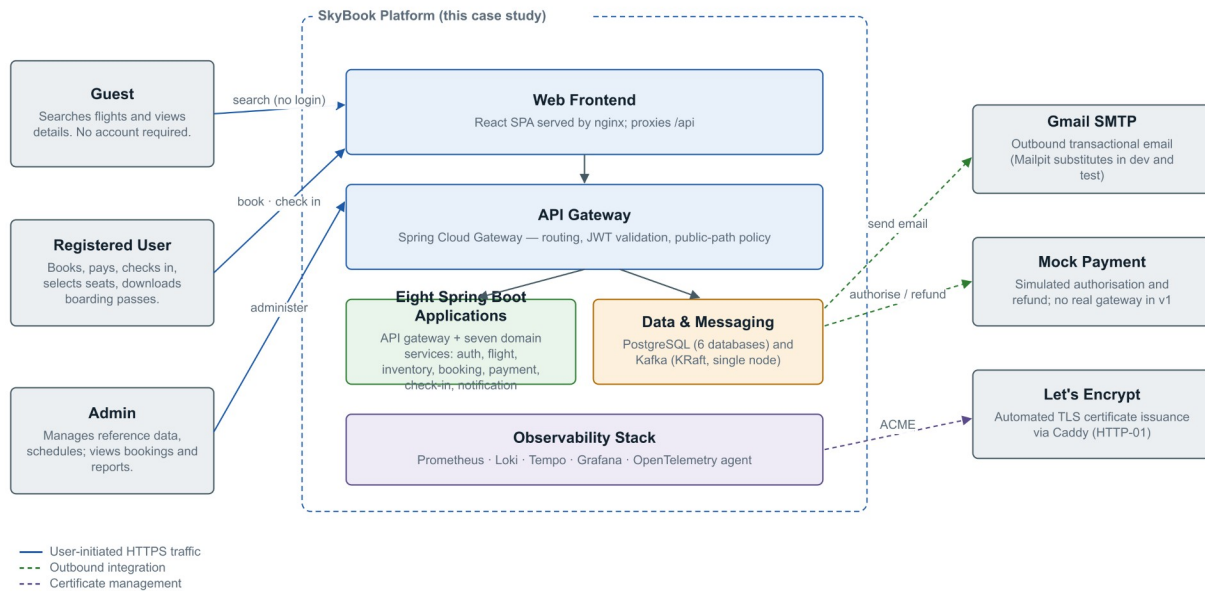
[Quality dashboard: sonarcloud.io/project/overview?id=praveencloudlab\\_skybook](https://sonarcloud.io/project/overview?id=praveencloudlab_skybook)

*Disclaimer: SkyBook is a personal portfolio project - not a real airline, affiliated with no carrier. Airline-style codes in seed data are illustration only; payments are simulated server-side and no cardholder data exists; all passenger records are synthetic.*

# The system

**Figure 1 — System Context**

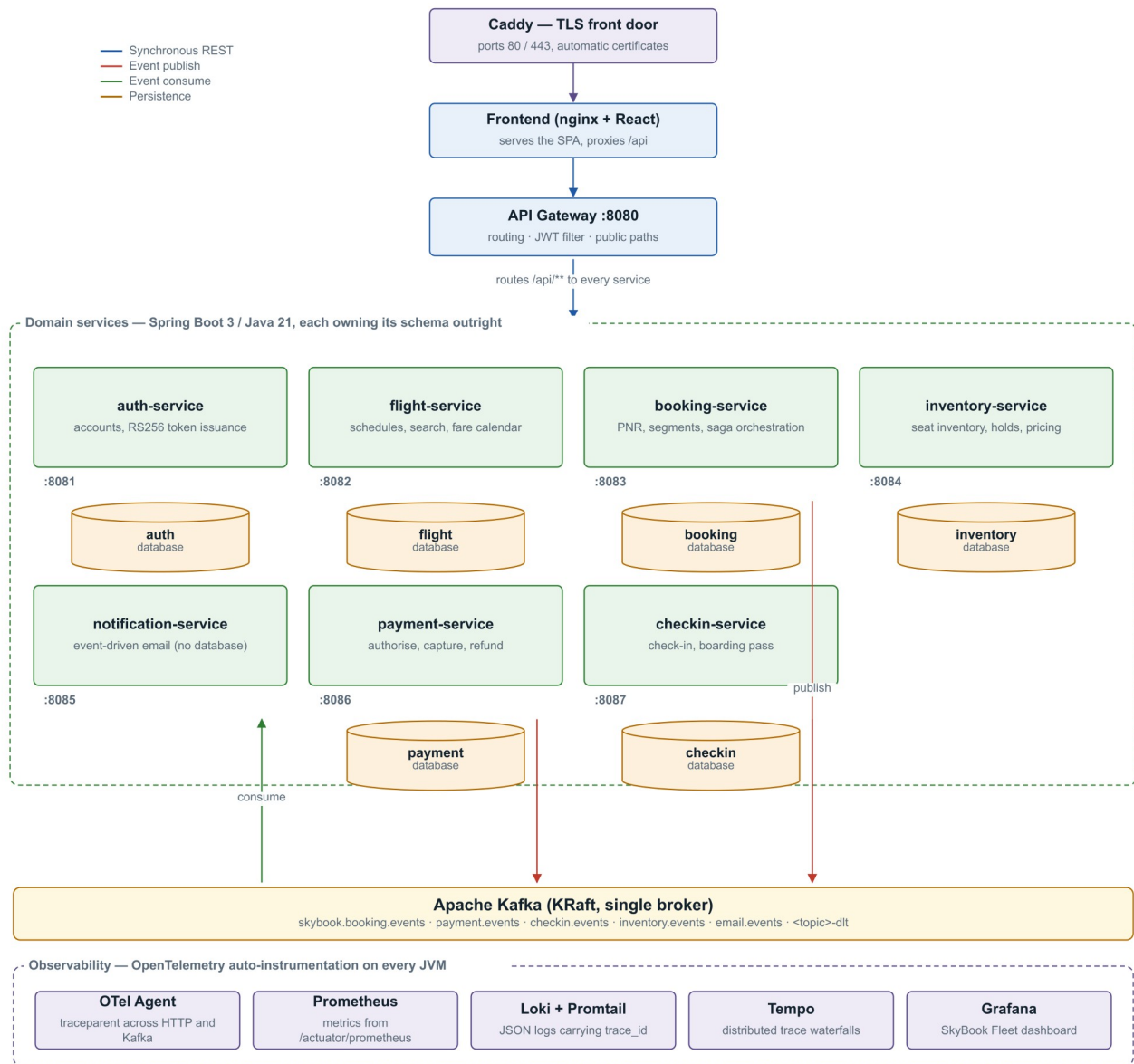
SkyBook passenger reservation platform and the parties around it



*The context: a public shopping surface, an authenticated booking core, and operations behind an admin role.*

**Figure 2 — Microservice Architecture**

Runtime components, synchronous routes and asynchronous event flow

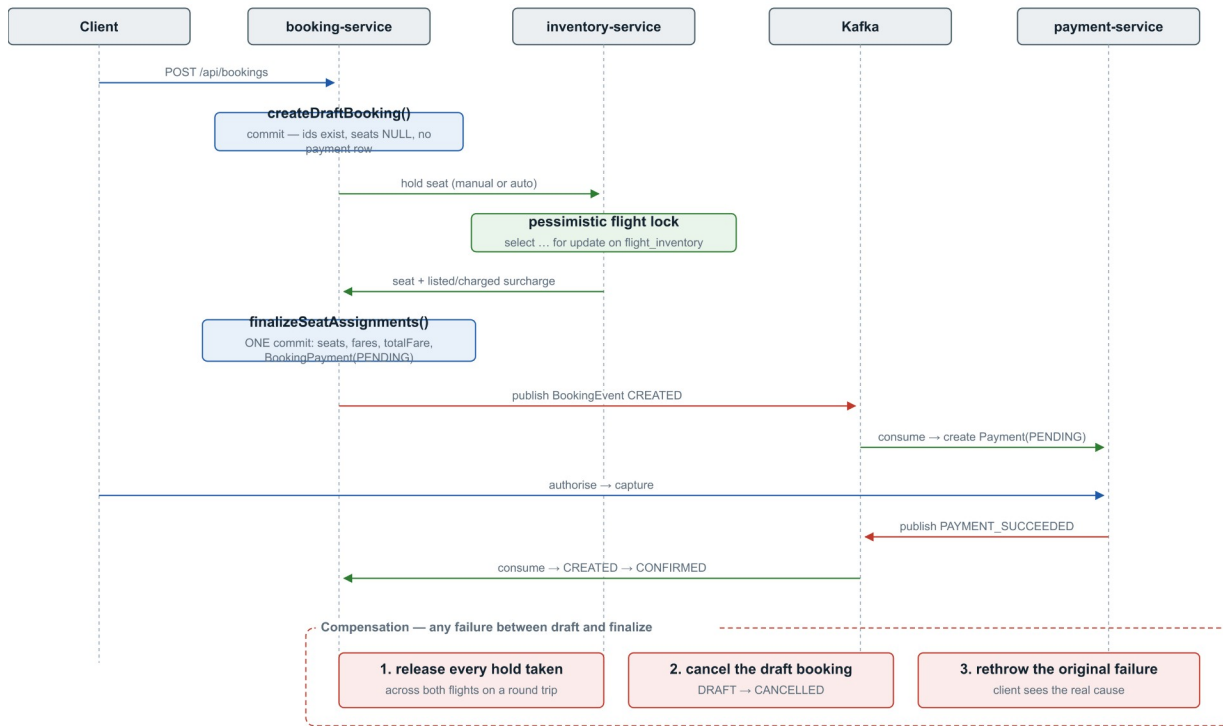


*Eight applications over one database instance (six schemas) and one event bus. Every service owns its data; nothing reaches another service's tables.*

## The hard part: money moves asynchronously

A booking is not confirmed when the payment call returns - it is confirmed when payment-service says so over Kafka and booking-service reacts. The saga below is the platform's spine, and the double-sell race (two real buyers, one seat, concurrent checkout) is certified end to end every night and on every promotion.

**Figure 5 — Booking Saga**  
draft → hold → finalize, with compensation on any failure

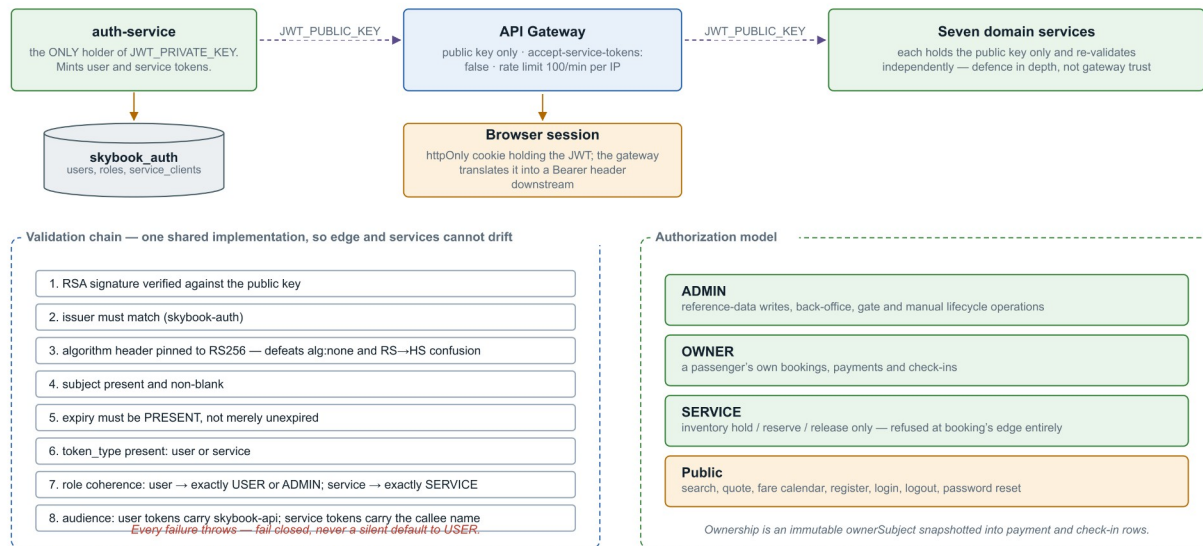


The facade is deliberately NOT @Transactional: every external call happens outside a database transaction, so no connection is held across network I/O.

The booking saga: draft, hold, pay, confirm - with compensations, not distributed locks.

## Security, stated honestly

**Figure 10 — Authentication and Authorization**  
RS256 asymmetric signing: one service mints, everyone verifies



*RS256 sessions, per-service credentials, an httpOnly cookie the gateway exchanges for a Bearer header - and the gaps stated next to the controls.*

The eight CSRF findings SonarCloud raised were closed with written justifications in code - the cookie is consumed at the gateway, SameSite=Lax covers the SPA, and all 48 GET endpoints were audited as read-only. A leaked mail credential early in the project was revoked, purged from history, and is recorded in the study as an incident, not hidden.

## Evidence, not adjectives

| Claim                        | Proof                                                                                                                                                      |
|------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1,722 tests, zero failures   | mvn clean verify at commit 96e24b2, 4 Aug 2026 - per-module table in the full study sums to exactly this                                                   |
| 91.2% coverage, gate passing | Public SonarCloud dashboard, measured on every push                                                                                                        |
| The journey certifies        | 24 e2e tests through the gateway only: booking, payment, check-in, boarding pass, double-sell race, email into a real sink, one trace across the Kafka hop |
| Fast enough under load       | k6: 3,405/3,405 checks, p95 search 60 ms, quote 26 ms, 0.00% errors - thresholds gate every promotion                                                      |
| Backups actually restore     | Weekly drill: back up, destroy the database volume, restore, verify 1,336,748 rows against the manifest - drilled live on 4 Aug                            |
| Same artifact everywhere     | Images built once per commit, scanned, published as multi-arch digests; every environment from DEV to PROD runs those exact bytes                          |

## The product

The screenshot displays the SkyBook flight search interface. At the top, there's a header with the SkyBook logo, language and currency selectors (English, £ GBP), and buttons for 'Search flights' and 'Sign in'. The main section is titled 'Search flights' with a subtitle: 'A year of real schedules across 30 routes. Compare fares, pick your seat from the actual cabin — no account needed to look.' Below this, a search bar allows users to select trip type (Round trip, One way, Multi-city), origin (London Heathrow (LH)), destination (Dubai (DXB)), and travel date (Tue 4 Aug). A 'Search' button is also present.

The results section shows a seven-day price strip for the route LHR → DXB on Tue 4 Aug. The prices are: 1 Aug (£136), 2 Aug (£136), 3 Aug (£136), 4 Aug (£136), 5 Aug (£136), 6 Aug (£136), and 7 Aug (£130.90). Below the price strip, there are 20 trips listed. The first three trips are shown in detail:

| Flight Type | Departure Time | Arrival Time | Duration | Airline | Fare (per person) |
|-------------|----------------|--------------|----------|---------|-------------------|
| Direct      | 08:25 LHR      | 15:20 DXB    | 6h 55m   | ER603   | £136.00           |
| Direct      | 06:53 LHR      | 13:53 DXB    | 7h       | S81679  | £136.00           |
| Direct      | 13:29 LHR      | 20:29 DXB    | 7h       |         | £136.00           |

Filters on the left include 'STOPS' (Direct, 1 stop, 2 stops) and 'DEPARTURE TIME' (Morning, Afternoon, Evening). The 'SORT BY' options are Earliest, Latest, and Shortest.

*Search: a seven-day price strip and one card per itinerary - priced by the same endpoint checkout will use.*

English

E GBP

Search flights

My trips

Profile

Sign out

Booking in progress: LHR → DXB 2026-08-05

Start a new search

LHR → DXB

Wed 05 Aug · 08:25

Back

Flights

Guests

Seats

Bags

Payment

Seat selection

Economy - Saver — secure your seat now, or let us assign one free at check-in.

Case Study

Free

Extra charge

Selected

Unavailable

Exit row

FRONT OF AIRCRAFT

|    |   |   |   |   |   |   |      |
|----|---|---|---|---|---|---|------|
|    | A | B | C | D | E | F |      |
| 15 | A | B | C | D | E | F |      |
| 16 | A | B | C | D | E | F |      |
| 17 | A | B | C | D | E | F |      |
| 18 | A | B | C | D | E | F |      |
| 19 | A | B | C | D | E | F |      |
| 20 | A | B | C | D | E | F | EXIT |
| 21 | A | B | C | D | E | F | EXIT |
| 22 | A | B | C | D | E | F |      |
| 23 | A | B | C | D | E | F |      |
| 24 | A | B | C | D | E | F |      |
| 25 | A | B | C | D | E | F |      |
| 26 | A | B | C | D | E | F |      |
| 27 | A | B | C | D | E | F |      |
| 28 | A | B | C | D | E | F |      |
| 29 | A | B | C | D | E | F |      |
| 30 | A | B | C | D | E | F |      |
| 31 | A | B | C | D | E | F |      |
| 32 | A | B | C | D | E | F |      |
| 33 | A | B | C | D | E | F |      |
| 34 | A | B | C | D | E | F |      |
| 35 | A | B | C | D | E | F |      |
| 36 | A | B | C | D | E | F |      |
| 37 | A | B | C | D | E | F |      |
| 38 | A | B | C | D | E | F |      |
| 39 | A | B | C | D | E | F |      |
| 40 | A | B | C | D | E | F |      |
| 41 | A | B | C | D | E | F |      |
| 42 | A | B | C | D | E | F |      |
| 43 | A | B | C | D | E | F |      |
| 44 | A | B | C | D | E | F |      |
| 45 | A | B | C | D | E | F |      |
| 46 | A | B | C | D | E | F |      |
| 47 | A | B | C | D | E | F |      |
| 48 | A | B | C | D | E | F |      |
| 49 | A | B | C | D | E | F |      |
| 50 | A | B | C | D | E | F |      |

BEAR OF AIRCRAFT

No seat chosen

Skip — assign me a seat (free)

Back

Add seats later

Continue

Summary

Flight details

Wed 05 Aug

08:25 — 18:20

LHR — DXB

6h 55m · Direct

Economy Saver

London Heathrow to Dubai

Guests

Mr Case Study

Fare summary

Total: £136.00

Including taxes

SkyBook

A working demonstration airline: real schedules, real seat maps, and a boarding pass you can actually scan.

BOOK

Search flights

My trips

ACCOUNT

Signed in as passenger.de...

Sign out

© 2026 SkyBook. A portfolio project — not a real airline.

30 routes · 16 airports · a year of departures

Seat selection from the real cabin: paid rows priced per seat, held seats disabled, free auto-assign.



 SkyBook

English ▾ £ GBP ▾ Search flights My trips Profile Sign out



Payment received

Your booking reference

SB4EFA

Paid£136.00

Payment referencePAY-2026-T887GQ

Seat19B

StatusCONFIRMED

E-ticket

Case Study125-0000023201

Confirmed. A confirmation email is on its way.

Done



A working demonstration airline: real schedules, real seat maps, and a boarding pass you can actually scan.

## BOOK

Search flights  
My trips

## ACCOUNT

Signed in as **passenger.de...**  
Sign out

*The moment three services agree: record locator, paid amount, trip saved.*

All trips

BOOKING REFERENCE  
**SB4EFA**  
Booked Tue 4 Aug

Confirmed

**£136.00**  
Download e-ticket

EK EK0079 Checked in

05:45

6h 50m  
Direct

09:35  
MAN

Passenger (1)

Seats 19B

Case Study  
Economy - Saver - seat 19B

**£136.00**

Total paid

**£136.00**

CONTACT

NAME  
Case Study

EMAIL  
passenger.demo@skybook.test

PHONE  
Not provided at booking

SkyBook

English

£ GBP

Search flights

My trips

Profile

Sign out

Case Study

125-0000023201 CT CHECKED\_IN

Manage booking

Cancellation & refund rules

- More than 72h before departure — full refund per your fare rules (Saver keeps its 30% fee; Flexi and Premium refund in full).
- 72h – 24h before departure — 50% of the fare-rule refund.
- Under 24h (same day) — cancellation still frees your seat, but the fare is not refunded.
- Last 2 hours before departure — online cancellation is closed; the airport desk can still help.
- Already checked in? — you can still cancel the whole booking; issued boarding passes are voided and the seats released.
- A refund due is returned automatically to the original payment method; check-in closes for every passenger.

Cancel booking...

Check-in & boarding passes

Case Study

EK0079 DXB → MAN Wed 5 Aug, 05:45 Seat 19B

Change seat checked in

Download boarding pass

SkyBook

BOARDING PASS

ECONOMY

PASSENGER  
CASE STUDY

BOOKING REF (PNR)  
SB4EFA

DXB  
Dubai - Terminal 3

MAN  
Manchester - Terminal 2

FLIGHT  
EK0079

DATE  
5 Aug 2026

DEPARTS  
05:45

BOARDING  
05:00

TERMINAL  
3

GATE  
TBA

SEAT  
19B

CABIN  
Economy

BOARDING GROUP  
4

NOTICE: Please arrive at the boarding gate by 04:30, 30 minutes before boarding begins at 05:00. The gate closes before departure and late passengers may be offloaded.

Issued 4 Aug 2026 11:04

BOARDING PASS

PASSENGER  
CASE STUDY

FLIGHT  
EK0079

DXB → MAN

SEAT  
19B

GSP  
4

BOARDS  
05:00

BP-2026-KPTGBR  
PNR SB4EFA

SkyBook

A working demonstration airline: real schedules, real seat maps, and a boarding pass you can actually scan.

BOOK

Search flights

My trips

ACCOUNT

Signed in as passenger.de...

Sign out

© 2026 SkyBook. A portfolio project — not a real airline.

30 routes · 16 airports · a year of departures

The boarding pass - the QR is real and the gate-operations console verifies it.

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## Five things that went wrong, on purpose in print

| Incident                                                                                         | Lesson                                                                                                                                 |
|--------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------|
| A transactional method silently ran with no transaction (Spring self-invocation)                 | Proxies do not apply to internal calls; the nightly schedule generator ran unprotected until static analysis flagged it                |
| Every cross-timezone arrival displayed hours wrong - durations looked right by accident          | Two wall clocks in different zones subtract to nonsense; fixing it touched generation, email, search, 414,679 rows and the screenshots |
| A round-trip cancellation could refund partially and leave the booking looking live              | The expansion set and the remaining set were computed from different keys; pinned by tests that fail on the old code                   |
| The environment ladder's first ever run refused a bad promotion, then caught its own seeding bug | Gates that only pass are decoration; these blocked, and the fixes are commits                                                          |
| A test demanded zero collisions from a generator whose own name promised "rare"                  | The birthday paradox made CI red one run in twenty-two; assert what the code promises, not more                                        |

Current limits, stated: no transactional outbox (a post-commit publish failure loses the event - top of the roadmap), no sustained capacity test (the k6 gate catches regressions, not limits), single-VM deployment with RPO 24h / RTO ~30min, and a browser UI suite not yet wired into CI. The full study gives every gap its reasons.

*The full 68-page engineering edition covers the architecture, the saga, pricing and refunds, check-in, the delivery history commit by commit, testing, defects and lessons, the data model, and six appendices - traceability, API contracts, security operations, screens, roadmap, glossary.*